Fiscal Year:	FY 2021	**Task Last Updated:**	FY 03/03/2021
PI Name:	Fernandez-Pello, Carlos Ph.D.		
Project Title:	Material Ignition and Suppression Test (MIST) in Space Exploration Atmospheres		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-1740	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	05/21/2013	**End Date:**	04/30/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
Contact Monitor:	Olson, Sandra	**Contact Phone:**	216-433-2859
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 4/30/2022 per NSSC information (Ed., 10/18/21) NOTE: End date changed to 10/31/2021 per NSSC information (Ed., 9/9/20) NOTE: Extended to 10/31/2020 per PI and S. Olson/GRC (Ed., 3/23/16) NOTE: Extended to 10/31/2015 per NSSC information (Ed., 3/24/15)		
Key Personnel Changes/Previous PI:	February 2021 report: Professor Michael Gollner, Co-PI.		
COI Name (Institution):	Gollner, Michael Ph.D. (University of California, Berkeley)		
Grant/Contract No.:	NNX13AL10A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of NNX10AE01G, Material Ignition and Suppression in Space Exploration Atmospheres, with the same principal investigator, Prof. Carlos Fernandez-Pello. Central Objectives: The next generation of NASA spacecraft will use lower pressures and increased oxygen concentrations than current designs. Although there is insufficient knowledge about fire behavior of materials in these Space Exploration Atmospheres (SEA), recent research suggests that material ignitability is increased in SEA. These results imply that next generation spacecraft present a greater fire safety hazard than current designs. The flammability, or fire hazard, of materials is normally assessed by the material's ease of ignition, rate of flame spread, heat released and extinction. In this research project, an experimental investigation is conducted to study the flammability of thermoplastic materials under SEA is being conducted. Methods/techniques: The proposed experimental apparatus is simple and easily incorporated in the Combustion Integrated Rack on the International Space Station (ISS). It consists of a small-scale cylindrical flow duct that has fuel samples attached to its inside wall. A cylindrical radiant heater is located at the center of the duct. An oxidizer gas flows at down the length of the tube. Electrical igniters are placed downstream of each fuel sample. Time to ignition and the time between introduction of suppression agent and extinction are measured as a function of environmental conditions. The normal gravity and microgravity experimental programs will be supported by a comprehensive model of piloted ignition and extinction of materials in SEA. Microgravity experiments will be used to validate the model, which in turn will be used to predict the fire behavior of different combustible materials in SEA. Perceived Significance: Currently, there exists no testing methodology specifically designed to determine the fire hazards of materials under those conditions. The proposed work will fill that void and provide additional information about the effect of SEA on the flammability and suppression of materials. It will also provide guidance to interpret and extend current NASA testing procedures to SEA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The project is intended to enhance the ability to predict ignition, burning (flame spread and mass burning), and extinction characteristics of materials in their end use configuration. This approach is called performance based design in terrestrial fire protection engineering. Computational support for this approach requires detailed information about material thermal and combustion properties. Our work on MIST will use experimental data to validate computer models attempting to predict fire behavior using material property estimation techniques. These techniques are currently a focus of research in fire protection engineering.
Task Progress:	Summary The opposed flame spread rate, flame appearance, and surface regression rate over polymethylmethacrylate (PMMA) cylinders has been studied under different normoxic conditions potentially applicable in future space exploration atmospheres (SEA). The environmental conditions were obtained by simultaneously reducing ambient pressure and increasing oxygen concentration. The flame spread rate is found to increase linearly with normoxic environments with high oxygen concentration, and with the mixed convection flow velocity that the flame is exposed to. It is also found that the flame spread rate depends on the rod diameter, with a transition to a stronger dependence as the normoxic conditions are changed to lower pressures and higher oxygen concentrations. This last result not only seems to contradict the predictions of analytical models of flame spread but also a phenomenological interpretation of the flame spread process. Similarly, the observation that the surface regression rate decreases as the normoxic conditions are changed to lower pressure and oxygen concentration appears to contradict phenomenological arguments of the process, indicating that cautions should be taken when applying simplified analyses of flame spread and surface regression to spacecraft environments. The modifications implemented in the normal gravity MIST apparatus to incorporate the radiant heaters will permit testing at different radiant fluxes to observe the effect of an external radiant flux on the flammability of materials in SEA. This testing will be initiated once the accurate operation of the modified apparatus is verified.
Bibliography Type:	Description: (Last Updated: 02/15/2026)
Articles in Peer-reviewed Journals	Thomsen M, Fernandez-Pello C, Olson S, Ferkul P. "Downward burning of PMMA cylinders: The effect of pressure and oxygen." Proceedings of the Combustion Institute. 2021 Jan 1;38(3):4837-44. https://doi.org/10.1016/j.proci.2020.05.024 , Jan-2021